EXERCISE (O-1)

Ideal Gas Laws: Boyle's Law

1. Which one of the following statements is not correct about the three states of matter i.e. solid, liquid and gaseous

(A) Molecules of a solid possess least energy whereas those of a gas possess highest energy

(B) The density of solid is highest whereas that of gases is lowest

(C) Gases, like liquids possess definite volumes

(D) Molecules of a solid possess vibratory motion

CSM091

2. Pressure of a gas in a vessel can be measured by

(A) Barometer (B) Manometer

(C) Stalgometer (D) All the above

CSM093

3. Which one of the following statements is wrong for gases

(A) Gases do not have a definite shape and volume

(B) Volume of the gas is equal to the volume of the container confining the gas

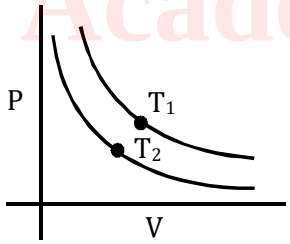
(C) Confined gas exerts uniform pressure on the walls of its container in all directions

(D) Mass of the gas cannot be determined by weighing a container in which it is enclosed

CSM094

4. According to graph, which is true :-

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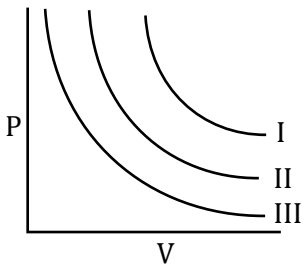


(A) $T_1 > T_2$ (B) $T_1 < T_2$

(C) $T_1 = T_2$ (D) None of these

CSM098

5. I, II, III are three isotherms respectively at T_1, T_2 and T_3 as shown in graph. Temperature will be inorder:



(A) $T_1 = T_2 = T_3$ (B) $T_1 < T_2 < T_3$

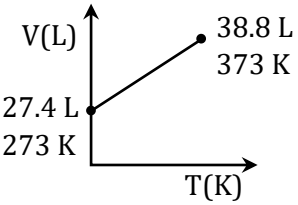
(C) $T_1 > T_2 > T_3$ (D) $T_1 > T_2 = T_3$

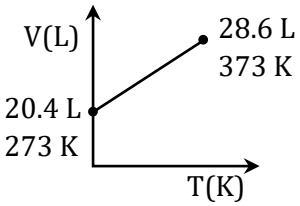
CSM101

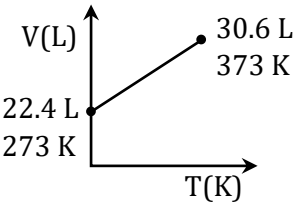
For More Join: @IITJEE_Advanced

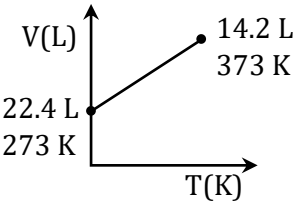
Charles' Law

6. Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atm ?

(A) 

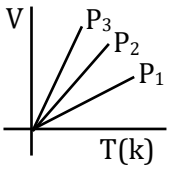
(B) 

(C) 

(D) 

CSM095

7. The volume-temperature graph of a given mass of an ideal gas at constant pressure are shown below. What is the correct order of pressures



(A) $P_1>P_3>P_2$ (B) $P_1>P_2>P_3$ (C) $P_2>P_3>P_1$ (D) $P_2>P_1>P_3$

CSM096

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8. 400 cm^3 of oxygen at 27°C were cooled to –3°C without change in pressure. The contraction in volume will be :-

(A) 40 cm^3 (B) 30 cm^3 (C) 44.4 cm^3 (D) 360 cm^3

CSM097

Dalton's Law of Partial Pressure

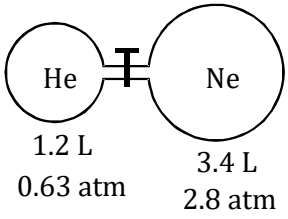
9. 56 g of nitrogen and 96 g of oxygen are mixed isothermally and at a total pressure of 10 atm. The partial pressures of oxygen and nitrogen (in atm) are respectively:-

(A) 4, 6 (B) 5, 5 (C) 2, 8 (D) 6, 4

CSM103

Container Problems

10. Consider the following apparatus. Calculate the partial pressure of helium after the opening valve. The temperature remains constant at 16°C.



(A) 0.164 atm (B) 1.64 atm (C) 0.328 atm (D) 1 atm

CSM104

For More Join: @IITJEE_Advanced

Graham's Law of Diffusion and Effusion

11. Which gas effuses fastest under identical conditions -

(A) N_2 (B) O_2 (C) Cl_2 (D) CH_4

CSM111

12. A gas X diffuses three times faster than another gas Y the ratio of their densities i.e., $D_x : D_y$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{9}$ (C) $\frac{1}{6}$ (D) $\frac{1}{12}$

CSM112

13. A 4 : 1 molar mixture of He & CH_4 kept in a vessel at 20 bar pressure. Due to a hole in the vessel, gas mixture leaks out. What is the composition (in mole ratio) of mixture effusing out initially -

(A) 8 : 1 (B) 4 : 1 (C) 1 : 4 (D) 4 : 3

CSM113

Kinetic Molecular Theory of Gases (KTG)

14. Four particles have speed 2, 3, 4 and 5 cm/s respectively. Their rms speed is :

(A) 3.5 cm/s (B) 3 cm/s (C) 4 cm/s (D) 3.67cm/s

CSM118

15. Temperature at which most probable speed of O_2 becomes equal to root mean square speed of N_2 is [Given : N_2 at 427°C]

(A) 732 K (B) 1200 K (C) 927 K (D) 800 K

CSM119

16. The density ratio of O_2 and H_2 is 16 : 1. The ratio of their U_{rms} at constant pressure is :-

(A) 4 : 1 (B) 16 : 1 (C) 1 : 4 (D) 1 : 16

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CSM120

17. The average speed of an ideal gas molecule at 27°C is 0.3 m/s. The average speed at 927°C will be -

(A) 0.6 m/s (B) 0.3 m/s (C) 0.9 m/s (D) 3.0 m/s

CSM121

Kinetic Energy and Molecular Speeds; Maxwell's distribution of molecular speed

18. If a gas expands at constant temperature then :

(A) No. of gaseous molecule decreases (B) kinetic energy of molecule decreases

(C) K.E. remain same (D) K.E. increases

CSM116

19. Average K.E. of CO_2 at 27°C is E. The average kinetic energy of N_2 at the same temperature will be

(A) E (B) 22E (C) E/22 (D) $E/\sqrt{2}$

CSM117

Collision theory: Z_1, Z_{11}, Z_{12} , Mean free path

20. Collision frequency (Z) of gas at a particular pressure.

(A) Decreases with the rise in temperature (B) Increases with the rise in temperature

(C) Decreases initially and then increases (D) Unpredictable

CSM123

21. At constant volume, Z_{11} is directly proportional to -

(A) $\sqrt{P}$ (B) P (C) T^2 (D) T

CSM124

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22. Calculate the mean free path in CO_2 at 27° C and a pressure of 10^{-9} bar.
[molecular diameter = 500 pm]
[Given: $R = \frac{25}{3}$ $Jmol^{-1}k^{-1}$; $\sqrt{2} = 1.4$; $\pi = \frac{22}{7}$; $N_A = 6 \times 10^{23}$]
(A) 3.78×10^5 cm (B) 3.78×10^4 cm (C) 3.78×10^3 cm (D) 6.78×10^3 cm

CSM125

Behaviour of Real Gases; Deviation from Ideal Gas Behaviour; Introduction of Van Der Waal's equation

23. Vander Waal's constant 'a' and 'b' are related with ..... respectively –
(A) Attractive force and bond energy of molecules
(B) Volume and repulsive force of molecules
(C) Shape and repulsive forces of molecules
(D) Attractive force and volume of the molecules

CSM134

24. In the Vander Waal's equation which will have highest value of 'a'.
(A) CH_4 (B) He (C) NH_3 (D) H_2

CSM135

25. In Vander Waal's equation of state of the gas law, the constant 'b' is a measure of :
(A) intermolecular repulsions (B) intermolecular attraction
(C) volume occupied by the molecules (D) intermolecular collisions per unit volume

CSM136

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26. Pressure of real gas is less than the pressure of ideal gas because :
(A) No. of collisions increases (B) Definite shape of molecule
(C) K.E. of molecule increases (D) Inter molecular forces

CSM137

27. The correct expression for the Vander Waals equation of state is :
(A) $\left(P + \frac{a}{n^2V^2}\right)(V - nb) = nRT$ (B) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = \Delta nRT$
(C) $\left(P + \frac{an^2}{V^2}\right)(V - b) = nRT$ (D) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

CSM138

28. At relatively high pressure, Vander Waals' equation reduces to :
(A) PV = RT (B) $PV = RT + \frac{a}{V}$
(C) PV = RT + Pb (D) $PV = RT - \frac{a}{V^2}$

CSM139

29. When does a real gas show behaviour same as ideal gas:
(A) At low temperature and low pressure (B) At high temperature and high pressure
(C) At low temperature and high pressure (D) At high temperature and low pressure

CSM140

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30. A real gas obeying Vander Waal's equation will resemble ideal gas, if the :

(A) constants a & b are small (B) a is large & b is small

(C) a is small & b is large (D) constant a & b are large

CSM141

31. If 'V' is actual volume of 1 molecule of gas then, excluded volume (b) of 1 mole of gaseous molecule is -

(A) $4\ N_AV$ (B) N_AV (C) V/N_A (D) V

CSM142

Compressibility Factor ; Graphs of Compressibility Factor

32. At low pressure the Vander Waals equation is reduced to -

(A) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RTV_m}$ (B) $Z = \frac{pV_m}{RT} = 1 + \frac{a}{RT}p$

(C) $pV_m = RT$ (D) $Z = \frac{pV_m}{RT} = 1 - \frac{a}{RT}$

CSM143

33. A gas obeys P(V – b) = RT. Which of the following are correct about this gas?

I. Isochoric curves have slope = $\frac{R}{V-b}$

II. Isobaric curves have slope $\frac{R}{P}$ and intercept b.

III. For the gas compressibility factor = $1 + \frac{Pb}{RT}$

IV. The attraction forces are overcome by repulsive forces.

(A) I (B) II, III (C) III (D) I, II, III, IV

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CSM144

34. Consider the equation $Z = \frac{PV}{RT}$, Which of the following statements is correct :

(A) When Z > 1 real gas are easier to compress

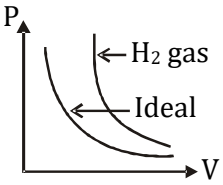
(B) When Z = 1 real gas are easier to compress

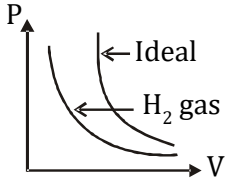
(C) When Z > 1 real gas are difficult to compress

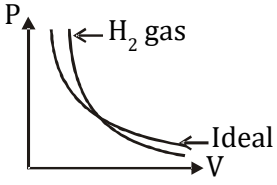
(D) When Z < 1 real gas are difficult to compress

CSM145

35. The correct graph to H_2 gas it :

(A) 

(B) 

(C) 

(D) All of these

CSM147

For More Join: @IITJEE_Advanced

36. The volume of gas A is twice than that of gas B. The compressibility factor of gas A is thrice than that of gas B at same temperature. The pressures of the gases for equal number of moles are :

(A) $2P_A = 3P_B$ (B) $P_A = 3P_B$

(C) $P_A = 2P_B$ (D) $3P_A = 2P_B$

CSM148

37. Compressibility of real gas will be less than ideal gas when

(T = temp. of gas T_b = boyle's temperature of gas)

(A) At very high pressure when $T > T_b$ (B) At very high pressure when $T < T_b$

(C) At low pressure when $T > T_b$ (D) At low pressure when $T < T_b$

CSM149

Virial Equation; Comparison with Vander Waal's Equation

38. The third virial coefficient of a He gas is 4×10^{-2} (lit/mol)^2, then what will be volume of 2 mole He gas at 1 atm 273K (273K > TB)

(A) 22.0 lit (B) 44.0 lit

(C) 44.8 lit (D) 45.3 lit

CSM146

Liquefaction; Boyle's Temperature; Graphs of Liquefaction of Real Gas (Andrew's Isotherm of CO_2)

39. Which gas can be easily liquefied?

given 'a' for $NH_3 = 4.17$, $CO_2 = 3.59$, $SO_2 = 6.71$, $Cl_2 = 6.49$

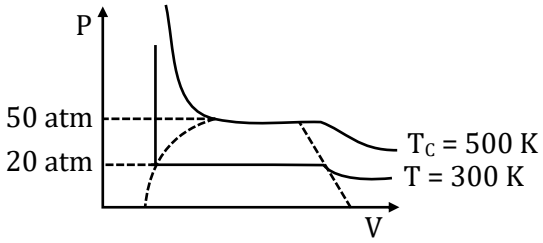
(A) NH_3 (B) Cl_2

(C) SO_2 (D) CO_2

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CSM153

40. For a real gas the P-V curve was experimentally plotted and it had the following appearance. With respect to liquefaction. Choose the correct statement.



(A) at T = 500 K, P = 40 atm, the state will be liquid.

(B) at T = 300 K, P = 50 atm, the state will be gas

(C) at T < 300 K, P > 20 atm, the state will be gas

(D) at 300 K < T < 500 K, P > 50 atm, the state will be liquid.

CSM159

41. The values of critical temperatures of following gasses are given:

Gases :	H_2	He	O_2	N_2
TC(K)	33.2	5.2	154.3	126

From the above data arrange the given gases in the increasing order of ease of their liquification.

(A) O_2, N_2, H_2, He (B) He, N_2, O_2, H_2

(C) He, H_2, N_2, O_2 (D) H_2, N_2, O_2, He

CSM156

For More Join: @IITJEE_Advanced

Critical Temperature; Critical Volume; Critical Pressure

42. From the below data what would be the decreasing order of liquefaction of these gases?

Gases :	H_2	N_2	NH_3	CO_2
Critical temperature :	33.2 K	126 K	405.5 K	304.1 K

(A) NH_3, CO_2, N_2, H_2 (B) CO_2, NH_3, H_2, N_2

(C) NH_3, N_2, H_2, CO_2 (D) H_2, N_2, CO_2, NH_3

CSM154

43. Liquefaction of H_2 and He is difficult, why ?

(A) Critical temperature are too low (B) Critical temperature are too high

(C) Statement is wrong (D) Only H_2 can be liquified

CSM155

44. If the critical constants for a hypothetical gas are V_C = 150 cm^3 mol^{-1}. P_C = 50 atm and T_C = 300K. Then the radius of the molecule is: [Take R = $\frac{1}{12}$ Ltr atm mol^{-1} K^{-1}]

(A) $\left(\frac{75}{2\pi N_A}\right)^{1/3}$ (B) $\left(\frac{75}{8\pi N_A}\right)^{1/3}$

(C) $\left(\frac{3}{\pi N_A}\right)^{1/3}$ (D) $\left(\frac{3}{256\pi N_A}\right)^{1/3}$

CSM158

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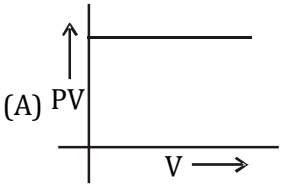
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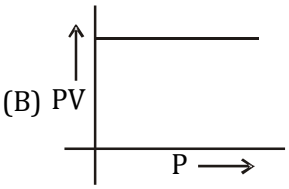
EXERCISE (O-2)

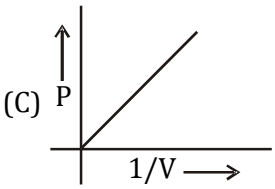
MULTIPLE CORRECT TYPE QUESTIONS

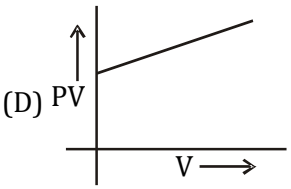
Ideal Gas Laws: Boyle's Law

1. For gaseous state at constant temperature which of the following plot is correct?

(A) 

(B) 

(C) 

(D) 

CSM105

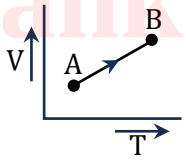
Ideal Gas Equation, Graphs Of Ideal Gas

2. Which of the following is/are CORRECT?

(A) At constant volume, for a definite quantity of an ideal gas, graph of PT v/s T^2 will be parabolic

(B) At constant pressure, for a definite quantity of an ideal gas, graph of VT v/s T^2 will be parabolic

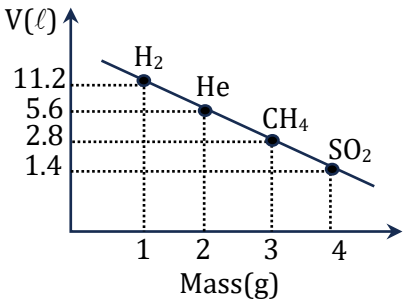
(C) In going from A to B for definite quantity of an ideal gas, pressure increase



(D) At constant volume, for a definite quantity of an ideal gas, graph of $\frac{P}{T}$ v/s T^2 will be straight line.

CSM324

3. At 1 atm and 273°C following is graphical representation of volume occupied by different gases. Which is not correctly matched?



(A) H_2 (B) He (C) CH_4 (D) SO_2

CSM325

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Dalton's Law of Partial Pressure

4. In a closed rigid container, 3 mol of gas A and 1 mol of gas B are mixed at constant temperature. If 1 mol of another gas C at same temperature is introduced and all gases are considered to be non-reacting, then

(A) Partial pressure of gases A and B remain unaffected due to introduction of gas C.

(B) Ratio of total pressure before and after mixing of gas 'C' is $\frac{3}{5}$.

(C) If the total pressure of gas mixture before introducing gas 'C' is 20 atm, then the total gas pressure after mixing 'C' will be 25 atm.

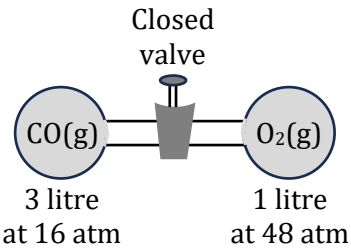
(D) If data of option 'C' are used, then partial pressure of gas 'C' will be 5 atm.

CSM106

Container Problems

5. Carbon mono oxide (CO) and oxygen O_2 react according to :

$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$



Assuming that the reaction takes place and goes to completion, after the valve is opened in the apparatus represented in the accompanying figure. Also assume that the temperature is fixed at 300 K. (Take R =0.08 atm L/mole K)

(A) Partial Pressure of O_2 = 6 atm.

(B) Number of moles of CO_2 formed = 2

(C) Number of moles of O_2 left = 1

(D) Partial Pressure of O_2 = 3 atm

CSM326

Collision theory: Z_1,Z_{11}, Z_{12} , Mean free path

6. Choose the correct statement(s) among the following -

(A) The mean free path (λ) of gaseous molecules is directly proportional to temperature of gas at constant volume

(B) The mean free path (λ) of gaseous molecules is inversely proportional to pressure of gas at constant volume

(C) The mean free path (λ) of gaseous molecules is directly proportional to volume of gas at constant T

(D) The mean free path (λ) of gaseous molecules is directly proportional to volume of gas at constant P

CSM131

7. One mole of helium and one mole of neon are taken in a vessel. Which of the following statements are correct?

(A) Molecules of helium strike the wall of vessel more frequently.

(B) Molecules of neon apply more average force per collision on the wall of vessel.

(C) Molecules of helium have greater average molecular speed.

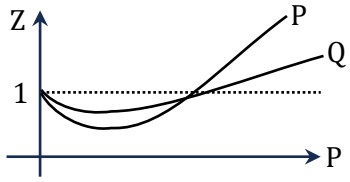
(D) Helium exerts higher pressure than neon.

CSM327

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Behaviour Of Real Gases, Deviation From Ideal Gas Behaviour, Introduction Of Van Der Waals Equation

8. Consider the following Z vs P graph for two difference real gases P and Q at the same temperature.



Which of the following statement(s) is/are correct?

(A) Gas P has greater attraction forces in lower pressure region

(B) Gas Q has greater repulsion forces in higher pressure region

(C) Gas P shows lesser deviation from ideal gas

(D) Gas Q shows lesser deviation from ideal gas

CSM328

9. Non-bonded interactions between two molecules consist of electrostatic (E) and van der Waals (VW) forces. Which is/are incorrect description(s) of distance dependence on intermolecular interaction between two molecules?

(A) At large distances, VW interaction decreases faster than E interactions

(B) At large distances E interaction decreases faster than VW interactions

(C) Both interactions have same distance dependence

(D) Which force drops faster depends on the net charge of the molecules only

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CSM329

Liquefaction; Boyle's Temperature; Graphs of Liquefaction of Real Gas (Andrew's Isotherm of CO_2)

10. Which of the following statements are correct ?

(A) It is not possible to compress a gas at a temperature below T_C

(B) At a temperature below T_C, the molecules are close enough for the attractive forces to act and condensation occurs

(C) No condensation takes place above T_C

(D) Boyle's temperature always greater than T_C

CSM160

Critical Temperature, Critical Volume, Critical Pressure

11. Which of following statement (s) is true

(A) Slope of isotherm at critical point is maximum.

(B) Larger is the value of T_C easier is the liquification of gas.

(C) Vander Waals equation of state is applicable below critical temperature at all pressure.

(D) All of the following

CSM330

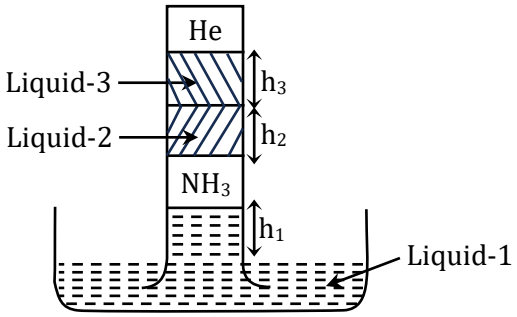
For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Ideal Gas Laws: Boyle's Law

Paragraph for Question # 12 to 13

In barometer system different liquids are placed along with trapped gases of NH_3 and He as shown in diagram. Use given data to answer following questions assuming temperature to remain constant at 300K. (Atmospheric pressure = 1 atm)



For liquid-1 $h_1 = 19$ cm $d_1 = 27.2$ g/cc
For liquid-2 $h_2 = 15.2$ cm $d_2 = 13.6$ g/cc
For liquid-3 $h_3 = 15.2$ cm $d_3 = 6.8$ g/cc

12. Pressure exerted by NH_3 gas will be:
(A) 0.4 atm (B) 0.5 atm (C) 0.75 atm (D) 0.2 atm

CSM331

13. Pressure exerted by He gas will be:
(A) 0.4 atm (B) 0.5 atm (C) 0.75 atm (D) 0.2 atm

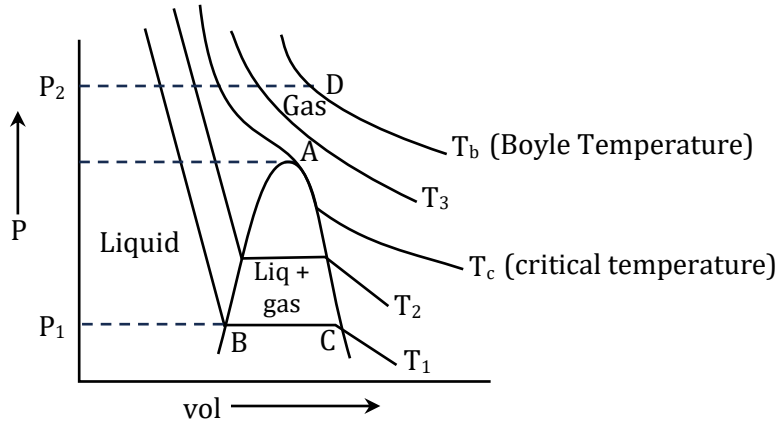
CSM332

Rankers Academy JEE

Liquefaction, Boyle's Temperature, Graphs Of Liquefaction Of Real Gas (Andrews Isotherm Of CO_2)

Paragraph for Question # 14

Andrews isotherm for CO_2 may be plotted as shown



14. At point 'A' , which of the following is true?
(P) compressibility factor becomes 0.375
(Q) the given gas is called vander Waals' gas
(R) below 'A' only one variable is required for liquefaction
(A) Only P, Q (B) Only Q, R (C) P, Q and R (D) Only P, R

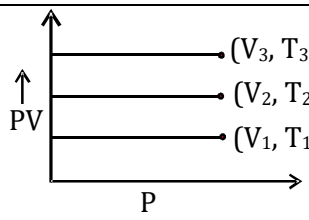
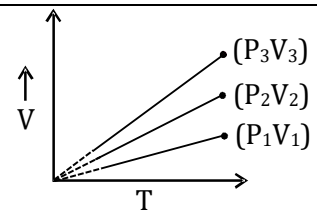
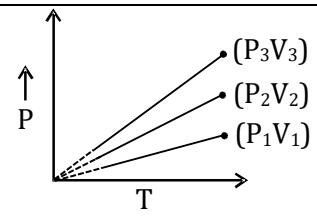
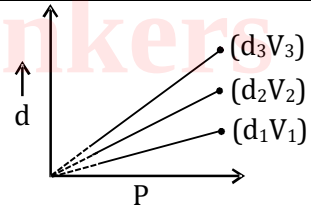
CSM333

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTIONS

Ideal Gas equation; Graphs of Ideal Gas

15. For a fixed amount of the gas match the two column :

	Column – I		Column – II
(A)		(p)	$T_1 > T_2 > T_3$
(B)		(q)	$P_1 > P_2 > P_3$
(C)		(r)	$V_1 > V_2 > V_3$
(D)		(s)	$d_1 > d_2 > d_3$

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(A) (A – q, s) ; (B - s) ; (C – p, r) ; (D - p)
(B) (A - s) ; (B - q, s) ; (C - r) ; (D - p, r)
(C) (A - r) ; (B - q, s) ; (C - q) ; (D - r, s)
(D) (A - q) ; (B - r, s) ; (C - r) ; (D - r, s)

CSM102

Kinetic Energy and Molecular Speeds; Maxwell's distribution of molecular speed

16. Match the following (where U_{rms} = root mean square speed, U_{av} = average speed, U_{mp} = most probable speed)

List I		List II	
(a)	U_{rms} / U_{av}	(i)	1.22
(b)	U_{av} / U_{mp}	(ii)	1.13
(c)	U_{rms} / U_{mp}	(iii)	1.08

(A) (a)-(iii), (b)-(ii), (c)-(i)
(B) (a)-(i), (b)-(ii), (c)-(iii)
(C) (a)-(iii), (b)-(i), (c)-(ii)
(D) (a)-(ii), (b)-(iii), (c)-(i)

CSM115

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

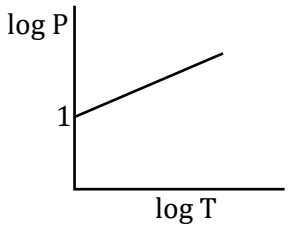
Ideal Gas Laws: Boyle's Law

1. A closed container containing O_2 and some liquid water was found to exert 740 mm pressure at 27°C. What will be the final pressure (in mm of Hg) if volume is reduced to half.
(consider volume of liquid water negligible, Given Aqueous tension of H_2O = 20 mm of Hg)

CSM109

Ideal Gas Equation, Graphs of Ideal Gas

2. In constant volume container of 0.821 litre, log P vs log T is plotted as shown in graph. Calculate number of moles of ideal gas present in container :

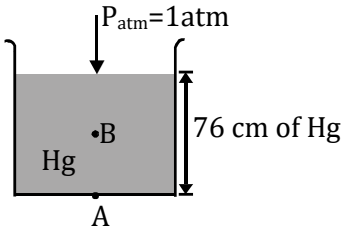


CSM110

Rankers Academy JEE

Container Problems

3. An open tank is filled with Hg upto a height of 76cm.



Find the pressure at the bottom (A) of the tank. (If atmospheric pressure is 1 atm)

CSM099

4. Two vessels whose volumes are in the ratio 2 : 1 contain nitrogen and oxygen at 2500 mm and 1000 mm pressures respectively when they are connected together what will be the pressure of the resulting mixture (in meters) ?

CSM107

Graham's Law of Diffusion And Effusion

5. Find the number of diffusion steps required to separate the isotopic mixture initially containing some amount of H_2 gas and 1 mol of D_2 gas in a container of 3 litre capacity maintained at 24.6 atm & 27 °C to the final mass ratio $\left(\frac{w_{D_2}}{w_{H_2}}\right)$ equal to $\frac{1}{4}$.

CSM174

For More Join: @IITJEE_Advanced

Collision Theory: Z_1,Z_{11}, Z_{12} , Mean Free Path

6. The number of bimolecular collision per cm^3 per second is Z_{11}. At constant temperature, by how much will Z_{11} change if the pressure is tripled in the vessel ?

CSM133

Behaviour Of Real Gases, Deviation From Ideal Gas Behaviour, Introduction Of Van Der Waals Equation

7. The Vander Waals constant for O_2 are a = 1.642 atm L^2 mol^{-2} and b = 0.04 L mol^{-1}. Calculate the temperature at which O_2 gas behaves ideally for longer range of pressure.

CSM151

8. Calculate molecular diameter for a gas if its molar excluded volume is 3.2 π ml. (in nanometer). Give the answer by multiplying with 10. (Take $N_A = 6.0 \times 10^{23}$)

CSM152

Liquefaction, Boyle's Temperature, Graphs Of Liquefaction Of Real Gas (Andrews Isotherm Of CO_2)

9. Critical constant of A gas

When pressure is increases at constant temp volume of gas decreases

AB $\rightarrow$ gases, BC $\rightarrow$ vapour + liquid, CD $\rightarrow$ liquid

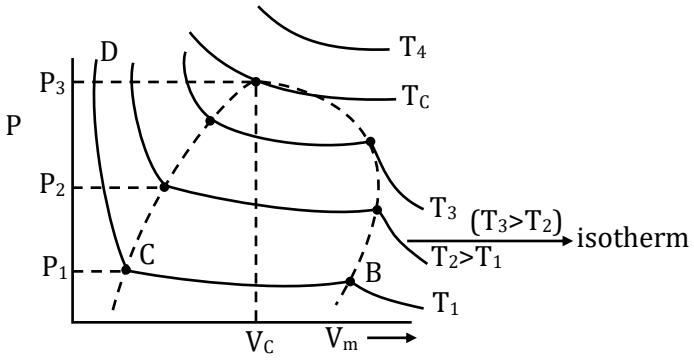
critical point : At this point all the physical properties of liquid phase will be same as the physical properties in vapour such as, density of liquid = density of vapour

T_C or critical temp : Temperature above which a gas cannot be liquified

P_C or critical pressure : minimum pressure which must be applied at critical temp to convert the gas into liquid.

V_C or critical volume : volume occupied by one mole of gas at T_C & P_C

Raikers' Academy JEE



If the ratio of PV_m & RT for a real gas is $\frac{x}{24}$ at a temp where $\left(\frac{\partial P}{\partial V_m}\right)$ = 0. The find value of 10x.

CSM161

10. Calculate the volume (in L) occupied by 0.2 mole of a Vander waal gas at 27°C and 0.082 atm.

[a = 4.10 L^2 atm mol^{-2}, b = $\frac{1}{6}$ $Lmol^{-1}$]

CSM181

For More Join: @IITJEE_Advanced

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Charles' Law

11. A gas occupies 100.0 mL at 50°C and 1 atm pressure. The gas is cooled at constant pressure so that volume is reduced to 50.0 mL. What is the final temperature of the gas.

CSM100

Ideal Gas Equation, Graphs Of Ideal Gas

12. A compound exists in the gaseous state both as a monomer (A) and dimer (A_2). The molecular weight of the monomer is 48. In an experiment, 96 g of the compound was confined in a vessel of volume 33.6 litres and heated to 273° C. Calculate the pressure (in atm) developed, if the compound exists as a dimer to the extent of 50 per cent by weight, under these conditions.

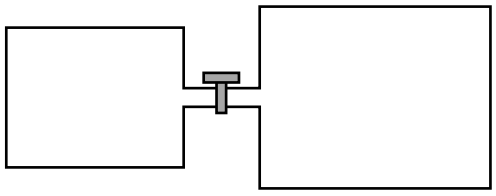
CSM168

13. To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to

CSM173

Rankers Academy JEE

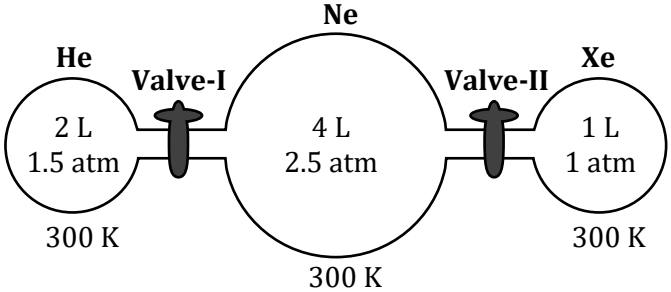
14. Two container (I) & (II) with volume 2 litre & 3 litre are separated through a stopcock as shown in the diagram. Container (I) contains three gases A, B, & C with mole ratio 1 : 2 : 3 and total pressure 30 atm. If stopcock is opened then, calculate the pressure (in atm) of gas C in container (II). (Assume temperature remains constant)



(I) (II)

CSM169

15. For the following diagram, calculate final pressure (in atm) when both valves are opened. Given that temperature (T) is same for each container.

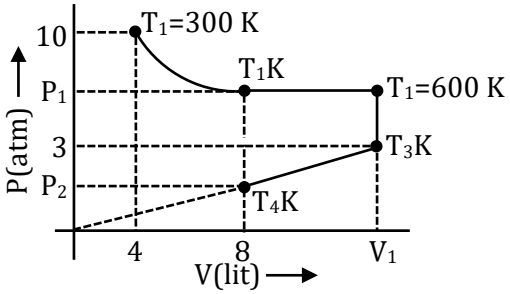


CSM170

For More Join: @IITJEE_Advanced

Ideal Gas Equation, Graphs of Ideal Gas

16. Fixed mass of a gas is subjected to the changes as shown is diagram, calculate T_3, T_4, P_1, P_2 and V_1 as shown is diagram. Considering gas obeys PV = nRT equation.



CSM164

17. A balloon containing 1 mole air at 1 atm initially is filled further with air till pressure increases to 3 atm. The initial diameter of the balloon is 1 m and the pressure at each state is proportion to diameter of the balloon. Calculate :

(a) No. of moles of air added to change the pressure from 1 atm to 3 atm.

(b) balloon will burst if either pressure increases to 7 atm or volume increases to 36p m3. Calculate the number of moles of air that must be added after initial condition to burst the balloon.

CSM165

18. One mole of an ideal gas is subjected to a process in which $P = \frac{1}{8.2} V$ where P is in atm & V in litre.

If the process is operating from 1 atm to finally 3 atm (no higher pressure achieved during the process) then what would be the maximum temperature (in °C) obtained.

CSM167

Rankers Academy JEE

Dalton's Law Of Partial Pressure

19. At 30°C dry air [75% N_2 + 25% O_2] is placed over H_2O (l) at 800 torr combined pressure of all 3 gases. If pressure is gradually increased isothermally to 1560 torr, then calculate partial pressure of O_2 at high pressure, in torr. [$V.P_{H_2O} = 40$ torr at 30°C].

CSM171

20. 16 gm of O_2 was filled in a container of capacity 8.2 lit. at 300 K. Calculate

(i) Pressure exerted by O_2 gas, in atm

(ii) Total pressure (in atm) of mixture of O_2 and O_3, if 50 % of oxygen is converted into ozone at same temperature.

(iii) Total pressure (in atm) exerted by gases, if 50% of oxygen is converted into ozone (O_3) at temperature 50 K.

CSM172

21. On a certain humid day, the mole fraction of water vapor in air at 25°C is 0.0287. If total pressure of air is 0.977 bar, calculate the partial pressure of water in air.

CSM108

Container Problems

22. One mole of NH_4Cl (s) is kept in an open container & then covered with a lid. The container is now heated to 600 K where all NH_4Cl (s) dissociates into NH_3 & HCl (g). The Volume of the container is 24.6 litres Assume that outside air is at 300 K and 1 atm pressure.

(i) What will be the final pressure (in atm) of gases inside the container.

(ii) Find whether the lid would stay or bounce off if it can with stand a pressure difference of 5.5 atm.

CSM166

For More Join: @IITJEE_Advanced

Graham's Law of Diffusion And Effusion

23. The rates of diffusion of two gases A and B are in the ratio 1 : 4. If the ratio of their masses present in the mixture is 2 : 3. The ratio of their mole fraction is : ($9^{1/3}$ = 2.08)

CSM122

Kinetic Energy And Molecular Speeds, Maxwell's Distribution Of Molecular Speed

24. The average speed of molecules in a sample of ozone gas is 1.84×10^4 cm s^{-1}. The average translational kinetic energy per molecules is xJ. The value of $\frac{x}{2} \times 10^{22}$ is

(Given : $N_A = 6 \times 1023$; $\sqrt{\frac{8}{3\pi}} = 0.92$)

CSM175

Collision Theory: Z_1,Z_{11}, Z_{12} , Mean Free Path

25. H_2 gas is kept inside a container A and container B each having volume 2 litre under different conditions which are described below. Determining the missing values with proper unit.
[R = 8 J mol^{-1} K^{-1} and $N_A = 6 \times 10^{23}$, N = No. of molecules]

Parameter	Container A	Container B
P	(i) ----------	1 atm
T	300 K	600 K
N	6×10^{20}	(ii) ----------
Total Average KE	(iii) ----------	(iv) ----------
Ratio U_{mps}	(v) ----------	
Ratio Z_{11}	(vi) ----------	

CSM176

Behaviour Of Real Gases, Deviation From Ideal Gas Behaviour, Introduction Of Van Der Waals Equation

Rankers Academy JEE

26. The density of mercury is 13.6 g/cm^3. Estimate the value of 'b' (in cm^3/ mole). (Hg = 200)

CSM177

27. Calculate the amount of He (in gm) present in the 10 litre container at 240 atm and 300K. Given value of "b" for He is 0.08 dm^3 mol^{-1}. (R = 0.08 atm lit mol^{-1} K^{-1})

CSM179

28. At what pressure (in atm) and 127°C, the density of O_2 gas becomes 1.6 g/L ?
[a = 4.0 atm L^2 mol^{-2}, b = 0.4 $Lmol^{-1}$, R = 0.08 L - atm/ K-mol]

CSM182

Compressibility Factor, Graphs of Compressibility Factor

29. The compression factor (compressibility factor) for one mole of a Vander Waals' gas at 0° C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the Van der Waals' constant 'a'.

CSM150

30. The molarity of O_2 gas at 72 atm and 300K is 6M. Calculate the value of Z for O_2.
(Use : R = 0.08 atm-litre/K-mole).

CSM178

Critical Temperature, Critical Volume, Critical Pressure

31. The critical temperature and critical pressure of a gas are 31°C and 728 atmospheres respectively. Calculate the constants 'a' and 'b'.

CSM162

32. For a real gas (mol. mass = 30) if density at critical point is 0.40 g/cm^3 and its $T_c = \frac{10^4}{41}$ K, then calculate Van der Waals constant a (in atm L^2mol^{-2}).

CSM180

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to

[JEE (Advanced) 2011]

CSM084

2. A closed tank has two compartments A and B, both filled with oxygen (assumed to be ideal gas). The partition separating the two compartments is fixed and is a perfect heat insulator (Figure 1). If the old partition is replaced by a new partition which can slide and conduct heat but does NOT allow the gas to leak across (Figure 2), the volume (in m^3) of the compartment A after the system attains equilibrium is____. **[JEE (Advanced) 2018]**

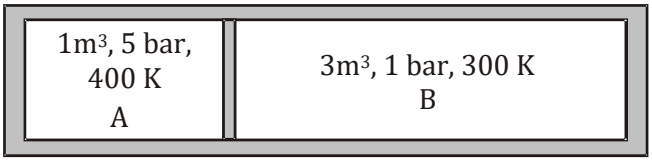


Figure 1

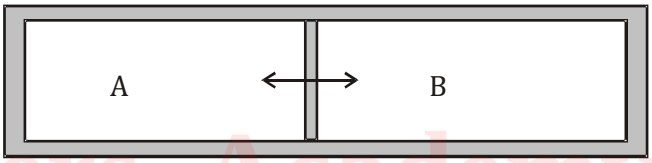


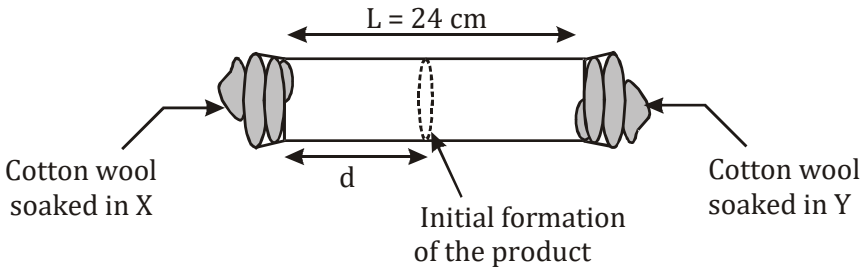
Figure 2

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CSM088

Paragraph for Question 3 & 4

X and Y are two volatile liquids with molar weights of 10g mol^{-1} and 40g mol^{-1} respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length L = 24 cm, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300K. Vapours of X and Y react to form a product which is first observed at a distance d cm from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours.



3. The value of d in cm (shown in the figure), as estimated from Graham's law, is –

[JEE (Advanced) 2014]

(A) 8 (B) 12 (C) 16 (D) 20

CSM086

For More Join: @IITJEE_Advanced

4. The experimental value of d is found to be smaller than the estimate obtained using Graham's law. This is due to – **[JEE (Advanced) 2014]**

(A) Larger mean free path for X as compared to that of Y

(B) Larger mean free path for Y as compared to that of X

(C) Increased collision frequency of Y with the inert gas as compared to that of X with the inert gas

(D) Increased collision frequency of X with the inert gas as compared to that of Y with the inert gas

CSM087

5. The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be : **[JEE (Advanced) 2005]**

(A) 4 (B) 2 (C) 1 (D) 0.5

CSM077

6. A closed vessel contains 10 g of an ideal gas **X** at 300 K, which exerts 2 atm pressure. At the same temperature, 80 g of another ideal gas **Y** is added to it and the pressure becomes 6 atm. The ratio of root mean square velocities of **X** and **Y** at 300 K is

[JEE (Advanced) 2024]

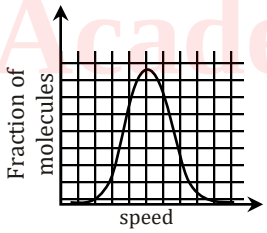
(A) $2\sqrt{2} : \sqrt{3}$ (B) $2\sqrt{2} : 1$ (C) $1 : 2$ (D) 2 : 1

CSM334

7. If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average and the roots mean square speeds, respectively, is

[JEE (Advanced) 2020]

Rankers Academy JEE



(A) 1 : 1 : 1 (B) 1 : 1 : 1.224 (C) 1 : 1.128 : 1.224 (D) 1 : 1.128 : 1

CSM090

8. Which of the following statement(s) is (are) correct regarding the root mean square speed (U_{rms}) and average translational kinetic energy (ε_{av}) of a molecule in a gas at equilibrium?

[JEE (Advanced) 2019]

(A) U_{rms} is doubled when its temperature is increased four times

(B) ε_{av} at a given temperature does not depend on its molecular mass

(C) U_{rms} is inversely proportional to the square root of its molecular mass

(D) ε_{av} is doubled when its temperature is increased four times

CSM089

9. At 400 K, the root mean square (rms) speed of a gas X (molecular weight = 40) is equal to the most probable speed of gas Y at 60 K. The molecular weight of the gas Y is **[JEE (Advanced) 2009]**

CSM083

10. The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature. **[JEE (Advanced) 2003]**

CSM075

For More Join: @IITJEE_Advanced

11. The root mean square velocity of an ideal gas at constant pressure varies with density as : **[JEE (Advanced) 2001]**

(A) d^2 (B) d (C) $d^{1/2}$ (D) $1/d^{1/2}$

CSM074

12. The pressure of a fixed amount of an ideal gas is proportional to its temperature. Frequency of collision and their impact both increase in proportion to the square root of temperature. True/False. **[JEE (Advanced) 2000]**

CSM073

13. The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of the gas : **[JEE (Advanced) 2000]**

(A) $T(H_2) = T(N_2)$ (B) $T(H_2) > T(N_2)$ (C) $T(H_2) < T(N_2)$ (D) $T(H_2) = \sqrt{7}\,T(N_2)$

CSM072

14. In thermodynamics the P-V work done is given by

$$w = -\int dV\, P_{ext}\ .$$

For a system undergoing a particular process, the work done is,

$$w = -\int dV\left(\frac{RT}{V-b} - \frac{a}{V^2}\right).$$

This equation is applicable to a **[JEE (Advanced) 2020]**

(A) System that satisfies the van der Waals equation of state.

(B) Process that is reversible and isothermal.

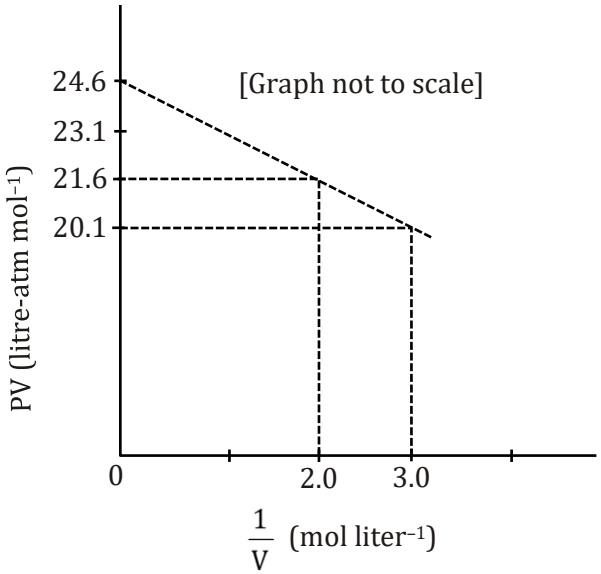
(C) Process that is reversible and adiabatic.

(D) Process that is irreversible and at constant pressure.

Rankers Academy JEE

CSM335

15. For one mole of a Vander Waals gas when b = 0 and T = 300 K, the PV vs. 1/V plot is shown below. The value of the Vander Waals constant a (atm. $liter^2\ mol^{-2}$) is **[JEE (Advanced) 2012]**



(A) 1.0 (B) 4.5 (C) 1.5 (D) 3.0

CSM085

For More Join: @IITJEE_Advanced

16. The term that corrects for the attractive forces present in a real gas in the Vander Waals equation is **[JEE (Advanced) 2009]**

(A) nb (B) $\frac{an^2}{V^2}$ (C) $-\frac{an^2}{V^2}$ (D) –nb

CSM082

17. A gas described by Vander Waals equation **[JEE (Advanced) 2008]**

(A) behaves similar to an ideal gas in the limit of large molar volumes

(B) behaves similar to an ideal gas in the limit of large pressures

(C) is characterised by Vander Waals coefficients that are dependent on the identity of the gas but are independent of the temperature

(D) has the pressure that is lower than the pressure exerted by the same gas behaving ideally

CSM081

18. Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

Column I	**Column II**
(A) Hydrogen gas (P = 200 atm, T = 273 K)	(P) Compressibility factor ≠ 1
(B) Hydrogen gas (P ~ 0, T = 273 K)	(Q) Attractive forces are dominant
(C) CO_2 (P = 1 atm, T = 273 K)	(R) PV = nRT
(D) Real gas with very large molar volume	(S) P (V– nb) = nRT

[JEE (Advanced) 2007]

(A) (A) → (P), (R); (B) → (P); (C) → (R), (Q); (D) → (Q)

(B) (A) → (Q), (S); (B) → (Q); (C) → (R), (S); (D) → (P)

(C) (A) → (R), (S); (B) → (S); (C) → (P), (R); (D) → (S)

(D) (A) → (P), (S); (B) → (R); (C) → (P), (Q); (D) → (R)

Pankers Academy JEE

CSM080

19. For a real gas obeying Vander Waal's equation, a graph is plotted between PV_m (y-axis) and P(x-axis) where V_m is molar volume. Find y-intercept of the graph. **[JEE (Advanced) 2005]**

CSM078

20. Positive deviation from ideal behaviour takes place because of. **[JEE (Advanced) 2003]**

(A) molecular interaction between atoms and $\frac{PV}{nRT} > 1$

(B) molecular interaction between atoms and $\frac{PV}{nRT} < 1$

(C) finite size of atoms and $\frac{PV}{nRT} > 1$

(D) finite size of atoms and $\frac{PV}{nRT} < 1$

CSM076

21. A gas has a compressibility factor of 0.5 and a molar volume of 0.4 $dm^3\ mol^{-1}$ at a temperature of 800 K and pressure **x** atm. If it shows ideal gas behaviour at the same temperature and pressure, the molar volume will be **y** $dm^3\ mol^{-1}$. The value of **x**/**y** is ___.

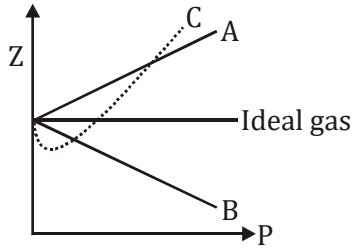
[Use: Gas constant, R = 8×10^{-2} L atm $K^{-1}\ mol^{-1}$]

[JEE (Advanced) 2023]

CSM336

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22.



where $Z = \frac{PV}{nRT}$,

a = Vander Waal's constant for pressure correction

b = Vander Waal's constant for volume correction

Pick the only incorrect statement **[JEE (Advanced) 2006]**

(A) for gas A, if a = 0, the compressibility factor is directly proportional to pressure

(B) for gas B, if b = 0, the compressibility factor is directly proportional to pressure

(C) for gas C, $a \neq 0$, $b \neq 0$, it can be used to calculate a and b by giving lowest P value and its intercept with Z = 1.

(D) slope for all three gases at high pressure (not shown in graph) is positive.

CSM079

23. The compressibility of a gas is less than unity at STP. Therefore : **[JEE (Advanced) 2000]**

(A) $V_m > 22.4$ L (B) $V_m < 22.4$ L

(C) $V_m = 22.4$ L (D) $V_m = 44.8$ L

CSM071

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	A	C	C	B	A	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	A	D	B	C	A	C	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	D	C	C	D	D	C	D	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	A	D	C	A	A	D	D	C	D
Que.	41	42	43	44						
Ans.	C	A	A	D						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABC	CD	ABCD	ACD	ABC	CD	ABC	AD	BCD	BCD
Que.	11	12	13	14	15	16				
Ans.	AB	B	D	C	B	A				

EXERCISE (S)

Rankers Academy JEE

1. 1460 **2.** 100 **3.** 2 **4.** 2 **5.** 4

6. 9 **7.** 500 **8.** 2 **9.** 9 **10.** 60

11. –111.5 **12.** 2.00 **13.** 7.00 **14.** 6.00 **15.** 2.00

16. (T_3 = 360.00, T_4 = 90.00, P_1 = 5.00, P_2 = 1.50, V_1 = 16.00) **17.** a. (80.00), b. (1295.00)

18. 627.00 **19.** 380.00 **20.** i. (1.50), ii. (1.25), iii. (0.20 or 0.21)

21. 0.028 **22.** i. (6.00), ii. (No) **23.** 0.347 **24.** 8.00

25. i. (0.012 atm), ii. (2.50×10^{22}), iii. (3.60 J), iv. (300.00 J), v. ($1 : \sqrt{2}$), vi. (1 : 2500)

26. 58.82 **27.** 222.22 **28.** 1.62 **29.** 1.256 **30.** 0.50

31. 0.36 & 0.00428 **32.** 1.68 or 1.69

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	7	2.22	C	D	B	D	B	ABC	4	434.17
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	T	C	ABC	C	B	ACD	D	RT	C
Que.	21	22	23							
Ans.	100	B	B							

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